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VARIETY SEEKING THROUGH BRAND SWITCHING

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A concept of variety seeking behavior is modeled as a stochastic brand choice model. The model yields a measure of variety seeking for each individual consumer. Panel data for 28 products are analyzed for each household and the possibility of market segmentation by variety seeking behavior is explored.

(Brand Choice Behavior; Variety Seeking; Segmentation)

Introduction

Observed brand choice behavior of consumers indicates that some consumers vary their consumption over many brands, while others avoid changes. Consumers who vary their consumption in some circumstances for some products may tend to conform to a very limited set of choices in others.

An extensive review of the literature on individual varied behavior leads McAlister and Pessemier (1982) to suggest a taxonomy of explanations for varied behavior. They dichotomize these explanations to "Derived" and "Direct" variation. "Derived varied behavior" refers to varied behavior that results from "forces that have nothing to do with a preference for change in and of itself" (McAlister and Pessemier 1982, p. 313). These forces are divided into "multiple needs" and "changes in the choice problem." Multiple needs include multiple users, contexts and uses. Laurent (1978) showed empirical evidence indicating that varied consumption of the household may result from different usage purposes of the product as well as different users within the household. Changes in the choice problem are changes in the feasible set of alternatives, tastes and constraints (new brands, advertising, deals, etc.)

"Direct variation" explanations of varied behavior rely on the "inherently satisfying aspects of changing behavior" (McAlister and Pessemier 1982, p. 314). They include interpersonal and intrapersonal motives. Interpersonal motives result from the needs for affiliation and distinction. Intrapersonal motives result from three main forces. First is the desire for the unfamiliar. McAlister and Pessemier (1982) cite empirical evidence on successful attempts to get stable and reliable measures of different aspects of this desire, that is related to an ideal level of stimulation desired by the individual.

Second is the desire for information, which was one of the factors of a scale used by Raju (1980) to measure optimum stimulation levels. Keon (1980) produced good predictions of brand choice probabilities by a model that rests on the assumption that

consumers want information on familiar brands. According to Keon, this need for information arises when continued consumption of a particular brand creates confusion with regard to the worth of other brands.

The third type of "direct variation" that satisfies intrapersonal needs is alternation among the familiar. McAlister and Pessemier (1982) cite studies in the areas of psychology and consumer behavior that show that levels of stimulation can be raised by switching among familiar as well as unfamiliar brands. Farquhar and Rao (1976) developed and tested a model in which varied consumption results from the attempt of the consumer to arrive at an optimal balance of product attributes. There is empirical evidence (McAlister and Pessemier 1982, p. 315) on the existence of ideal levels of attributes wanted by consumers in their consumption (also called satiation levels). This fact may result in switching among familiar brands that are rich in different attributes. McAlister (1979) used the notion of attribute satiation to model a choice collection of brands in a point of time. She further developed her model (McAlister 1982) to incorporate the dynamic aspect of consumption over time, and applied it to experimental data. This extended model relates attributes inventories to satiation levels in order to account for the dependence of brand choice on the history of brand choices made by a consumer.

A general model that attempts to account for a substantial part of varied consumption should include operational measures of all the psychological constructs that were mentioned above, as well as the situational variables that were grouped under the general heading of "derived variation." The interest in this study is limited to the phenomenon of alternation among familiar brands due to intrapersonal motives. As shown by Keon (1980), these may include the desire for information. Thus, it is not intended to predict varied consumption which is a result of many factors, but rather to shed some light on one aspect of this phenomenon.

Although there is substantial experimental evidence to support intrapersonal explanations for variety seeking behavior, there are no empirical findings to indicate how widespread this phenomenon is and what is its intensity. It is the purpose of this study to investigate these aspects of varied consumption. As will be shown later, variety seeking behavior has implications for the branding policy of the firm. The ability to measure consumers' variety seeking tendencies in a certain product market will bring about better understanding of the brand switching in this market. Moreover, knowledge of its intensity will help the firm improve its competitive position.

The present study interprets varied consumption as a result of variety seeking behavior. It is operationalized as a measure of individual tendency to vary consumption. This tendency is measured on a continuum that extends from extreme tendency to vary consumption to an extreme tendency to avoid variety.

In the following sections brand switching behavior is modeled as a variety seeking phenomenon. The model is fitted to panel data for individual households and some attempts of segmenting the households by their variety seeking measures are described. In a section that deals with possible extensions, the basic model is modified to deal with product market partitioning.

The empirical results show a great deal of heterogeneity among households as well as among products.

Development of the Model

Definitions and Assumptions

The notion of variety seeking (or avoidance) behavior is defined here to be the phenomenon of an individual consumer switching brands (or repeat buying) induced

by the utility (or disutility) she derives from the change itself, irrespective of the brands she switches to or from.¹ This behavior is idiosyncratic to the individual in her consumption of brands within a product class. Thus it is possible for an individual consumer to seek varied consumption in one product class and try to avoid variety in another product class. Brand selection behavior is determined not only by the utility (or disutility) derived from switching brands, i.e. mere change, but also by some underlying preferences for different brands. Thus, the observed sequence of brands selected by an individual consumer is a result of a combination of two forces: (a) the utility derived from the consumption of a specific brand and (b) the utility (or disutility) derived from switching by itself. It is important to notice that the definition above does not constrain the consumer's utility derived from switching brands to be nonnegative. Thus, variety seeking is viewed as a continuum ranging from extreme tendency for change to an extreme tendency to avoid change. Obviously, between the two extreme positions there is a point of indifference toward change, i.e., no change is actively sought or rejected.

The underlying preferences of a consumer for different brands in the product class are modeled here as a relative utility measure according to the Luce Choice Axiom (1959) in the same way it was done in previous work on brand switching behavior (Bass, Jeuland and Wright 1976). Thus, if a consumer has a certain nonnegative utility, u_i , associated with consumption of brand i , and if there are n brands in her evoked set for the product class, then her preference for brand i is $\theta_i = u_i / \sum_{i=1}^n u_i$. θ_i is often interpreted as the probability of selecting brand i in the context of a multinomial zero-order model of brand switching behavior (e.g., Bass, Jeuland and Wright 1976).

The Structure of the Model

The definition and assumptions given above have specific implications on the mathematical structure of brand switching behavior. With preferences θ_i being constant over time and purchase occasions, the selection of a brand in any particular purchase occasion depends on the brands selected in previous purchase occasions only to the extent to which variety is sought by the consumer. Furthermore, the definition of variety seeking as being induced by utility derived from change by itself implies that the next choice of a brand depends on the last choice only. This is simply because a change can occur only between two consecutive purchase occasions. Thus, a change is a termination of one run and a beginning of another run. (Run is a sequence of purchase occasions on which the same brand was bought.) The utility for an individual from brand i is assumed here to be a combination of two components, one is brand specific and the other is situation specific, i.e., it depends on the brand chosen in the last choice occasion. The specific functional form chosen here is a simple weighting of these two components. One mathematical formulation that can be used to describe this form of variety seeking is a first order Markov process. It was chosen for its simplicity and widespread use in brand choice modelling.

The probability of a consumer choosing brand j on the next purchase occasion will therefore depend on the brand chosen at the last purchase occasion. But since the utility is derived from the change itself irrespective of the brands involved, the probability of choosing a brand will depend on whether this brand was chosen on the last purchase occasion. Thus, the probability of switching to brand j will not depend on the brand switched from, i.e., $P_{ij} = P_{kj}$ for all $i, k \neq j$, where P_{ij} is the probability of choosing brand j given that brand i was chosen in the last purchase occasion. It was assumed, however, that the choice of a brand is determined by some combination of brand preference, θ_i , and variety seeking.

¹This is the simplest assumption possible. It is also very restrictive and it will be relaxed later.

Suppose for example that the behavior of a consumer is governed by an extreme search for variety with no regard to brand preference. Then her probability of buying the same brand in two consecutive times would be zero, i.e., $P_{jj} = 0$. Her probability of choosing a brand other than the one chosen on the last time will be the same for all brands, since no regard to preference is assumed, i.e., $P_{ij} = 1/(n-1)$ for all $i \neq j$, where n is the number of brands in the consumer's evoked set. Suppose further that another consumer has no utility (positive or negative) from active search of variety. In this case her probabilities of switching and repeat buying will depend only on the preferences for the brands, $P_{ij} = \theta_j$ and $P_{jj} = \theta_j$. A consumer who considers variety and brand preference in her choice may combine the two elements by assigning to them some weights, to get the intermediate case of variety seeking

$$P_{ij} = V \cdot \frac{1}{n-1} + (1-V)\theta_j \quad \text{for } i \neq j, \quad (1)$$

$$P_{jj} = V \cdot 0 + (1-V)\theta_j = (1-V)\theta_j, \quad (2)$$

where $1 \geq V \geq 0$. Thus, $V = 1$ indicates behavior that is guided strictly by variety seeking and $V = 0$ indicates disregard to variety or satisfaction with the variety gained by chance. V , therefore, can be used as a measure of the degree of active variety seeking.

The model for consumers who resist change and prefer inertia can be derived in a similar way. Suppose a consumer has no regard to brand preference and tries to avoid change at all costs. Her probability of switching brands is zero: $P_{ij} = 0$, and her probability of repeat buying is one: $P_{jj} = 1$. Suppose another consumer has no utility (positive or negative) from change. Her probability of switching brands and repeat buying are $P_{ij} = \theta_j$ and $P_{jj} = \theta_j$, respectively, as before. A consumer who combines inertia with brand preference considerations will assign weights to the two forces to get

$$P_{ij} = I \cdot 0 + (1-I)\theta_j = (1-I)\theta_j \quad \text{for } i \neq j, \quad (3)$$

$$P_{jj} = I \cdot 1 + (1-I)\theta_j = I + (1-I)\theta_j, \quad (4)$$

where $1 \geq I \geq 0$. Thus $I = 1$ indicates behavior that is guided strictly by inertia, and $I = 0$ is identical to $V = 0$ above. The model in (3) and (4) was suggested by Jeuland (1979) who termed it an INERTIA MODEL. Let $VS = V$ for $1 \geq V \geq 0$, and $VS = -I$ for $1 \geq I \geq 0$ to get the complete model of variety seeking:

$$P_{ij} = \frac{|VS| + VS}{2(n-1)} + (1-|VS|)\theta_j, \quad (5)^2$$

$$P_{jj} = \frac{|VS| - VS}{2} + (1-|VS|)\theta_j. \quad (6)$$

Thus, VS is a measure of active variety seeking which ranges from extreme search for variety ($VS = 1$) to extreme rejection of variety ($VS = -1$).

Table 1 gives the probability of switching to a brand, P_{ij} , for different levels of variety seeking, VS , and different values of relative preference, θ_j , in a five brands market ($n = 5$).

Equations (5) and (6) together with Table 1 show two important differences in the behavior of variety seekers and avoiders. First, the choice probabilities of variety

²I owe this specific formulation to an anonymous referee.

TABLE 1
Probability of Switching to Brand j in Five Brands Market

θ_j	$VS = -1.0$	$VS = -0.5$	$VS = 0.0$	$VS = 0.5$	$VS = 1.0$
0.00	0.	0.	0.	0.125	0.25
0.20	0.	0.100	0.20	0.225	0.25
0.25	0.	0.125	0.25	0.250	0.25
0.50	0.	0.250	0.50	0.375	0.25
0.75	0.	0.375	0.75	0.500	0.25
1.00	0.	0.500	1.00	0.625	0.25

avoiders ($VS < 0$) do not depend on the number of brands in the market, while they do so for variety seekers ($VS > 0$). This difference is not surprising since the variety seeking force influences only the probability of staying with the same brand for avoiders, while it influences only the probability of switching for seekers. This effect can be seen best for $VS = 1$ in Table 1, in which case $P_{ij} = 1/(n - 1)$, which implies that as the market selection (n) goes up, the probability of switching to any particular brand is going down. The higher the degree of variety seeking (VS) the more pronounced this effect is, and when no variety is desired ($VS \leq 0$) the effect vanishes altogether.

The second noticeable difference is that for variety seekers the probability of switching to a brand is pulled away from its relative preference (θ_j) towards an even distribution of the probability across all brands that were not bought in the last purchase occasion. Thus the effect of variety seeking is to lower the probability of switching to a high preference brand and to increase that of a low preference brand. In contrast, the effect of avoiding variety ($VS < 0$) is always to lower the probability of switching to a brand as compared with the seekers and the indifferent ($VS = 0$).

In order to evaluate the importance of this difference the transition probabilities of two consumers, a seeker and an avoider, can be compared. Let the superscripts $+$ and $-$ indicate the parameters of a variety seeker ($VS > 0$) and a variety avoider ($VS < 0$), respectively. Consider two consumers having VS coefficients of the same magnitude but of opposite signs, $VS^+ = |VS^-| = VS$. In this case if their relative preferences to a brand were the same, $\theta_j^+ = \theta_j^-$, then the switching probability to brand j of the variety seeker would be larger than that of the variety avoider, $P_{ij}^+ > P_{ij}^-$. In contrast, the repeat purchase probability of the seeker would be smaller than that of the avoider, $P_{jj}^+ < P_{jj}^-$. In the model presented above, all the attributes that are brand specific are summarized in the relative preferences θ_j 's. Thus, the effect of any change in these attributes, including those changes that result from competitive actions, will be directly reflected in the θ_j 's. The final effect, however, will be determined by the value of the consumer's variety seeking coefficient VS . It is therefore reasonable to ask what should be the magnitude of the difference $\theta_j^+ - \theta_j^-$ that will equate the repeat or the switching probabilities of avoiders and seekers. In fact, $\theta_j^+ - \theta_j^-$ can be looked upon as the difference between the drawing power of a brand for seekers and avoiders necessary to equate their switching or repeat probabilities. This difference can be obtained by equating the corresponding probabilities in (1) to (4). For the switching probabilities, $P_{ij}^+ = P_{ij}^-$ implies

$$VS^+/(n - 1) + (1 - VS^+)\theta_j^+ = (1 + VS^-)\theta_j^-.$$

Assuming $VS^+ = |VS^-| = VS$ and simplifying yields

$$\theta_j^- - \theta_j^+ = VS/(1 - VS)(n - 1).$$

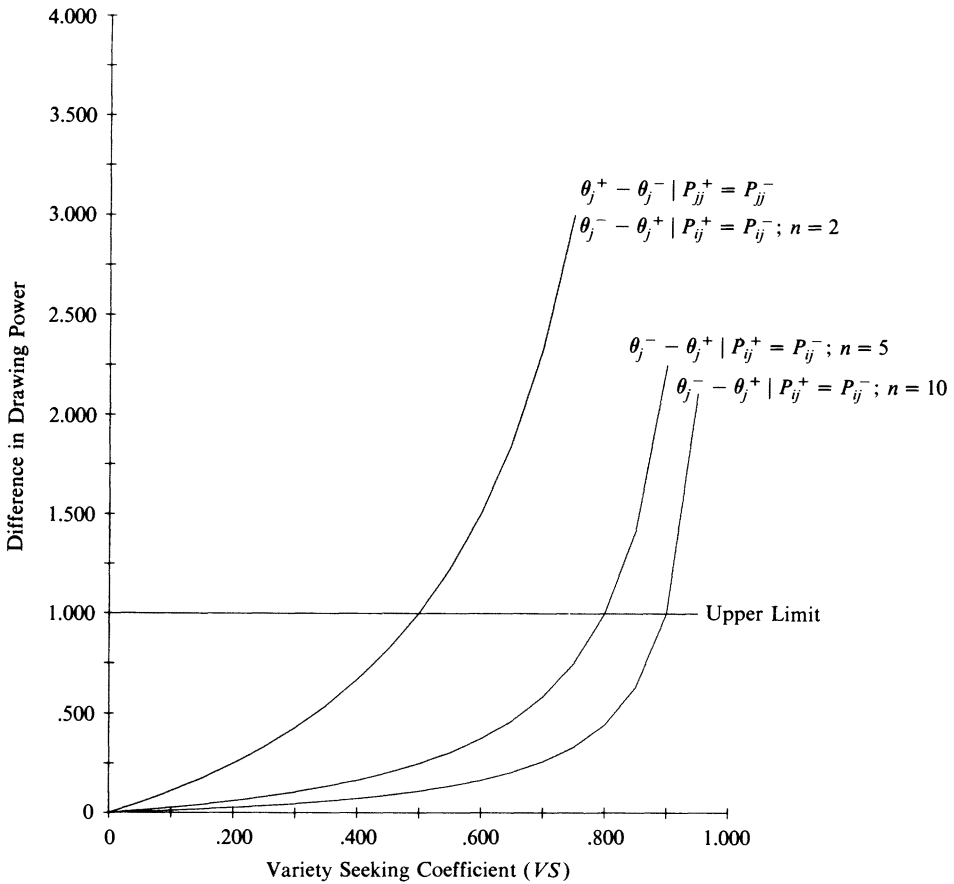


FIGURE 1. Comparison of Drawing Power for Variety Seekers and Avoiders.

For the repeat probabilities $P_{jj}^+ = P_{jj}^-$ implies

$$(1 - VS^+) \theta_j^+ = -VS^- + (1 + VS^-) \theta_j^-.$$

Assuming $VS^+ = |VS^-| = VS$ and simplifying yields

$$\theta_j^+ - \theta_j^- = VS / (1 - VS).$$

These differences are shown in Figure 1 as a function of VS and the number of brands, n . The horizontal line at 1.0 represents the upper limit of this difference. It shows that for $|VS| > 0.5$ it is impossible to equate the repeat probabilities of a seeker and an avoider. Likewise it is impossible to equate the switching probabilities of a seeker and an avoider having $|VS|$ larger than 0.5, 0.8, and 0.9 when the number of brands is 2, 5 and 10 respectively. If one accepts the idea that relative preference, θ_j , is at least partially the result of marketing effort, then the graphs in Figure 1 show how much more effort should be devoted to persuade the variety avoiders to switch to a brand, over and above the effort necessary to achieve the same results for variety seeker. The graph labeled $\theta_j^+ - \theta_j^- | P_{jj}^+ = P_{jj}^-$ shows how much more effort should be devoted to retain the seeker over and above the effort necessary to retain the avoider. This latter difference does not depend on the number of brands in the market, but the former does. As can be seen from Figure 1 the necessary difference in drawing power decreases as the number of brands increases. In fact, this difference can be ignored when the number of brands is very large, and the only important difference is that necessary to retain consumers with the same brand over successive purchases. Thus in

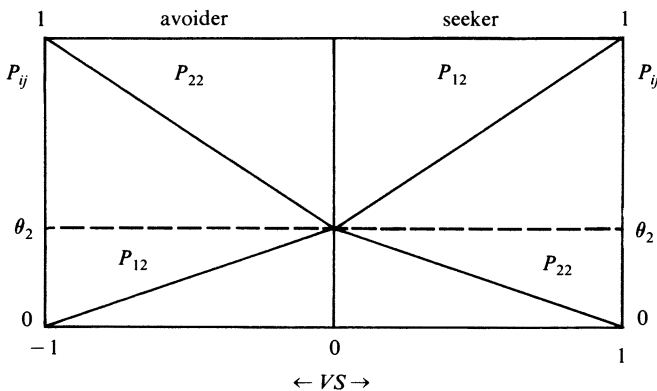


FIGURE 2. Variety Seeking in a Two-Brand Market.

a market with many brands it may be as easy to persuade the avoiders to switch to a brand as it is with the seekers, but it will still be much harder, if at all possible to maintain the seeker than the avoider. When dealing with a product for which variety seeking is important the problem of maintaining consumers and market share may be best dealt with by periodical introduction of new brands that will take advantage of the variety switching in the market.

In the special case of two-brands market the model in (1) to (4) becomes

$$\left. \begin{aligned} P_{12} &= VS + (1 - VS)\theta_2 = \theta_2 + (1 - \theta_2)VS, \\ P_{22} &= (1 - VS)\theta_2 = \theta_2 - \theta_2 VS, \end{aligned} \right\} \quad 1 \geq VS \geq 0, \quad (1a) \quad (2a)$$

$$\left. \begin{aligned} P_{12} &= (1 + VS)\theta_2 = \theta_2 + \theta_2 VS, \\ P_{22} &= -VS + (1 + VS)\theta_2 = \theta_2 - (1 - \theta_2)VS, \end{aligned} \right\} \quad 0 \geq VS \geq -1. \quad (3a) \quad (4a)$$

This case is also depicted in Figure 2. It is evident from Figure 2 that in the two-brands case the brand switching behavior of the variety avoider is the mirror image of the behavior of the variety seeker. The probability of switching to brand 2 increases with the degree of variety seeking from zero to θ_2 , at a rate of θ_2 , for avoiders, and from θ_2 to 1, at a rate of $(1 - \theta_2)$, for seekers. Correspondingly P_{22} goes down at a rate of θ_2 until it reaches θ_2 , and then at a rate of $(1 - \theta_2)$ to zero. At the point of variety indifference ($VS = 0$) the probability of choosing a brand is equal to its relative preference.

Comparison with Other Models

Three first-order Markeov models of brand choice behavior were suggested in the literature. All three models assume two-brands market (or one brand vs. a group of all other brands) and impose a specific structure on the transition probabilities. Morrison (1966) suggested the brand loyal model (BL) and the last purchase loyal model (LPL). Jones (1973) suggested another model (JM) within the wider context of a heterogeneous composite model. The transition matrices of the three models are

BL	LPL	JM
$2 \begin{bmatrix} p & 1-p \\ kp & 1-kp \end{bmatrix}$	$2 \begin{bmatrix} p & 1-p \\ 1-kp & kp \end{bmatrix}$	$2 \begin{bmatrix} \alpha + \beta & 1 - \alpha - \beta \\ \alpha & 1 - \alpha \end{bmatrix}$
$1 \geq p, k \geq 0$	$1 \geq p, k \geq 0$	$\alpha, \beta \geq 0; \alpha + \beta \leq 1.$

Since all three models have two parameters each, they can be compared to the variety

TABLE 2
Comparison of VS Model with Other Models

		$VS = 1$	$1 > VS > 0$	$VS = 0$	$0 > VS > -1$	$VS = -1$
	k	θ_1/θ_2	θ_1/θ_2	θ_1/θ_2	$\frac{1 - \theta_2(1 + VS)}{1 - \theta_1(1 + VS)}$	1
LPL	p	0	$\theta_2(1 - VS)$	θ_2	$1 - \theta_1(1 + VS)$	1
	kp	0	$\theta_1(1 - VS)$	θ_1	$1 - \theta_2(1 + VS)$	1
	k	∞	$\frac{1 - \theta_1(1 - VS)}{\theta_2(1 - VS)}$	1	$\frac{\theta_2(1 + VS)}{1 - \theta_1(1 + VS)}$	0
BL	p	0	$\theta_2(1 - VS)$	θ_2	$1 - \theta_1(1 + VS)$	1
	kp	1	$\theta_2 + \theta_1 VS$	θ_2	$\theta_2(1 + VS)$	0
JM	α	1	$\theta_2 + \theta_1 VS$	θ_2	$\theta_2(1 + VS)$	0
	β	-1	-VS	0	-VS	1

seeking model (*VS*) in (1a) to (4a) by solving for their parameters in terms of *VS*, θ_2 and $\theta_1 = 1 - \theta_2$. A summary of this comparison is given in Table 2. It is evident from Table 2 that the concepts of loyalty that are embodied in the parameter *k* of the BL and LPL models are combinations of the preference for variety and the relative preferences of the brands. Furthermore, because of the constraint imposed on *k* to be no larger than one the BL and LPL models cannot capture the whole range of brand choice behavior that is captured by the *VS* model. The JM is also constrained to the case of variety avoidance (*VS* < 1) by the requirement that β be nonnegative. Thus, the *VS* model can be thought of as an unconstrained, multiple brands generalization of the other three models. Finally, all four models can accommodate a zero-order brand choice behavior. But while for the BL, LPL and JM models the zero-order case is an anchor point on the boundary of their constraints, it is a mid-point for the *VS* model. Thus, the zero-order model represents an extreme behavior for the former and a moderate behavior for the latter.

Estimation and Empirical Results

Estimation

The general form of the log-likelihood function of a chronological sequence of brand choices of an individual consumer, given a first-order Markov model, can be written as

$$\log F(n_{ij} | P_1, P_{ij}) = \log P_1 + \sum_i n_{ii} \log P_{ii} + \sum_i \sum_{j \neq i} n_{ij} \log P_{ij} \quad \text{where:} \quad (7)$$

P_1 is the equilibrium probability of the first brand in the sequence;³
 n_{ij} is the number of transitions from brand *i* to brand *j*, *i* and *j* = 1, 2, . . . , *n*, the number of distinct brands in the sequence;
 P_{ij} is the transition probability as defined before.
The model in (5) and (6) specifies the probabilities P_{ij} in terms of the parameters θ_i , *VS* and *n*. Substituting these into the general form in (7) yields the log-likelihood

³For *VS* > 0

$$P_1 = \frac{VS/(n - 1) + (1 - VS)\theta_1}{1 + VS/(n - 1)},$$

and for *VS* < 0

$$P_1 = \theta_1.$$

functions of a sequence of brand choices given the values of the parameters

$$\begin{aligned} \log F(n_{ij} | \theta_i, n, VS > 0) &= \log P_1 + \sum_i n_{ii} \log [\theta_i (1 - VS)] \\ &+ \sum_i \sum_{j \neq i} n_{ij} \log [VS / (n - 1) + (1 - VS) \theta_j], \end{aligned} \quad (7a)$$

$$\begin{aligned} \log F(n_{ij} | \theta_i, n, VS < 0) &= \log P_1 + \sum_i n_{ii} \log [\theta_i (1 + VS) - VS] \\ &+ \sum_i \sum_{j \neq i} n_{ij} \log [\theta_j (1 + VS)], \end{aligned} \quad (7b)$$

$$\log F(n_{ij} | \theta_i, n, VS = 0) = \sum_i n_i \log \theta_i, \quad (7c)$$

where n_i is the number of times brand i appears in the sequence. Maximum likelihood estimates of VS and the θ_i 's for an individual consumer's sequence can be obtained by maximizing the log-likelihood functions in (7a) and (7b) with respect to VS and the θ_i 's, subject to the model's constraints, $\sum_i \theta_i = 1$; $1 \geq VS \geq 0$ for (7a) and $-1 \leq VS \leq 0$ for (7b). The first-order conditions for this maximization imply the solution for the maximum likelihood estimates $\hat{\theta}_i$,

$$\hat{\theta}_i = \frac{n_i - BVS / (n - 1) + [(n_i - BVS / (n - 1))^2 + 4Bn_{ii}VS / (n - 1)]^{1/2}}{2B(1 - VS)}, \quad VS > 0, \quad (8a)$$

$$\hat{\theta}_i = \frac{(1 + VS)n_i + VSA + [(VSA + (1 + VS)n_i)^2 + 4(n_{ii} - n_i)VS(1 + VS)A]^{1/2}}{2(1 + VS)A}, \quad VS < 0, \quad (8b)^4$$

$$\hat{\theta}_i = \frac{n_i}{N}, \quad VS = 0, \quad (8c)$$

where N is the total number of purchases in the sequence, $A = N + VS(N - 1)$ and $B = N - VS / (n - 1 + VS)$.

Since $\hat{\theta}_i$ was specified in terms of VS a complete solution can be obtained by finding an estimate $\widehat{VS}$ that will satisfy the constraint $\sum_i \hat{\theta}_i = 1$. One such solution is obviously $\widehat{VS} = 0$, as (8c) is a special case of both (8a) and (8b). Thus, the search is for a second solution $\widehat{VS} \neq 0$ that will yield a log-likelihood function (7a) or (7b) larger than (7c). If the resulting likelihood function is not significantly larger than that of the special case, $VS = 0$ (the zero order multinomial model) then for this particular consumer indifference toward variety is a "good enough" approximation. The asymptotic likelihood ratio test can be used to make probability statements on the magnitude of VS . Thus, the random variable

$$-2 \log [F(n_{ij} | \hat{\theta}_i, n, VS = 0) / F(n_{ij} | \hat{\theta}_i, n, \widehat{VS} \neq 0)]$$

is asymptotically distributed as χ^2 with one degree of freedom.

From the discussion above, it may seem as if the two parts of the model, viz.,

⁴The likelihood function and the solution of $\hat{\theta}_i$ for $VS < 0$ are given by Jeuland (1979), as this case is identical to his INERTIA MODEL.

$VS > 0$ and $VS < 0$ (and their corresponding likelihood functions) are in fact two separate, possibly rival, models. Billingsley (1961, Theorem 2) shows that this is not the case, and the model in (5) and (6) has a unique solution. Indeed, in no case out of the many that were tried, nonzero solutions were obtained for *both* $VS > 0$ and $VS < 0$. In other words, when a nonzero solution for $\widehat{VS}$ maximizes (7a) a zero solution, $\widehat{VS} = 0$ maximizes (7b) and vice versa.

A computer program was used to search for $\widehat{VS}$, that satisfies $\sum_i \hat{\theta}_i = 1$, over the zero-one interval. The algorithm employed combines brute-force procedure with the Powell algorithm (see e.g., Kuester and Mize 1973).

Empirical Results

The data base for the empirical analysis includes consumers' diary information from three panels of consumers: (1) Chicago Tribune Panel, 1958–66, (2) Chicago Tribune Panel, 1968, (3) Market Research Corporation of America (MRCA) panel, 1962–66. Empirical results were obtained for 28 frequently bought consumer products: six products from the first panel, twenty-one from the second and one product from the third. Only households with 30 or more purchases from the first panel and 20 or more purchases from the second and third panels were included in the analysis.

Maximum likelihood estimates of θ_i 's and VS were obtained for each individual household, within each product class following the procedure that was described in the previous section. Next, the null hypothesis that $VS = 0$ was tested with the likelihood ratio test and all of the households for which the null hypothesis cannot be rejected can be treated as being indifferent to change ($VS = 0$), i.e., following the zero-order process.

Before conducting the test a specific significance level (α) should be chosen. Since the author had no strong prior beliefs one way or the other it seemed reasonable to choose α in such a way that on average the probabilities of making errors of both types will be equal, i.e. $\alpha = \beta$. See Sawyer and Ball (1981) for an extensive discussion on statistical power in marketing research.

Following the procedure described in the Appendix, β probabilities (one minus the power of the test) were computed for each consumer for each combination of five α levels and ten VS_1 values under H_1 . Within each product class results were averaged over the consumers, and standard deviations from these means were obtained. The resulting β means and standard deviations for Regular Coffee and Salad Dressing are given in Table 3. These two products were chosen as examples since they represent two extreme types of behavior. The mean $\widehat{VS}$ coefficient for Regular Coffee and Salad Dressing were -0.10 and 0.19 , respectively. 46% of the consumers of Regular Coffee had $\widehat{VS} < 0$ and 18% had $\widehat{VS} > 0$ (excluding brand loyals), while 12% and 60% were the corresponding figures for Salad Dressing.

The results for β in Table 3 show that if one is interested only in large deviations from the null hypothesis, i.e., $|VS_1| \geq 0.30$, the traditional significance levels of 0.01 or 0.05 are appropriate for maintaining the balance between the confidence and the power of the test. However, if it is important to discriminate between $VS_0 = 0$ and $|VS_1| = 0.20$ an α level of 0.10 is needed, and for $|VS_1| = 0.10$ significance level as high as 0.25 may be necessary to equate α and β . It is interesting to compare the results for the two product classes in this respect. For Salad Dressing, where the majority of consumers had zero or positive $\widehat{VS}$, the probability of erroneously accepting the null hypothesis ($VS_0 = 0$) when in fact $\widehat{VS} = -0.10$ is low and $\alpha = 0.10$ may be appropriate. However, when the estimate $\widehat{VS}$ is close to the alternative hypothesis ($VS_1 = -0.10$), as was the case for many consumers of Regular Coffee, the discrimination power

TABLE 3
Means and Standard Deviations of β Probabilities for Likelihood Ratio Test

a. Regular Coffee		Variety Seeking Coefficient under H_1									
α		- 0.10	- 0.20	- 0.30	- 0.40	- 0.50	0.10	0.20	0.30	0.40	0.50
0.01	Means	0.63	0.29	0.12	0.04	0.01	0.75	0.35	0.13	0.04	0.00
	Standard Deviation	0.27	0.28	0.18	0.08	0.02	0.21	0.25	0.16	0.07	0.01
0.05	Means	0.45	0.16	0.06	0.02	0.00	0.56	0.20	0.06	0.01	0.00
	Standard Deviation	0.26	0.19	0.10	0.03	0.01	0.22	0.17	0.08	0.03	0.00
0.10	Means	0.36	0.11	0.03	0.01	0.00	0.46	0.13	0.03	0.01	0.00
	Standard Deviation	0.24	0.15	0.06	0.02	0.00	0.21	0.13	0.06	0.01	0.00
0.25	Means	0.22	0.06	0.01	0.00	0.00	0.29	0.06	0.01	0.00	0.00
	Standard Deviation	0.18	0.08	0.03	0.01	0.00	0.16	0.08	0.03	0.01	0.00
b. Salad Dressing		Variety Seeking Coefficient under H_1									
α		- 0.10	- 0.20	- 0.30	- 0.40	- 0.50	0.10	0.20	0.30	0.40	0.50
0.01	Means	0.24	0.04	0.01	0.00	0.00	0.77	0.22	0.03	0.01	0.00
	Standard Deviation	0.24	0.11	0.04	0.01	0.00	0.18	0.21	0.09	0.04	0.01
0.05	Means	0.13	0.02	0.00	0.00	0.00	0.58	0.11	0.01	0.00	0.00
	Standard Deviation	0.17	0.06	0.02	0.00	0.00	0.20	0.14	0.05	0.02	0.00
0.10	Means	0.09	0.01	0.00	0.00	0.00	0.47	0.07	0.01	0.00	0.00
	Standard Deviation	0.13	0.04	0.01	0.00	0.00	0.19	0.10	0.04	0.01	0.00
0.25	Means	0.04	0.00	0.00	0.00	0.00	0.30	0.03	0.00	0.00	0.00
	Standard Deviation	0.08	0.02	0.00	0.00	0.00	0.16	0.06	0.02	0.00	0.00

of the test is low and a 0.25 significance level is needed to bring the power to par with the confidence level.

The importance of the magnitude of VS was analyzed in a previous section and is depicted in Figure 1. It has been shown that small deviations of VS from zero may not be so crucial in drawing a consumer into a specific brand (P_{ij}), especially when there are many brands in the market. However, VS as small as 0.10 may be of great importance when trying to maintain a consumer with the same brand over purchase occasions (P_{jj}). Combining these results with those on the power of the test in Table 3, a significance level of $\alpha = 0.25$ was chosen for the likelihood ratio test. The distributions of the estimated variety seeking coefficients for the 28 products are given in Table 4.

The “Brand Loyal” column of Table 4 gives the percentage of households in the sample that purchased only one brand of the respective product in the time period of analysis. This behavior is consistent with two different sets of values of the parameters of the variety seeking model in (5) and (6). Let j indicate the only brand that was purchased by the “brand loyal” household. Then $VS = 0$; $\theta_j = 1$ and $\theta_i = 0$ for all $i \neq j$ yield $P_{ij} = 1$ and $P_{jj} = 1$. Thus, since the household started its sequence with brand j it will always repeat this choice. This interpretation of absolute loyalty implies that the household is indifferent toward variety (no feedback effect) and has absolutely no utility from any other brand but brand j : $\theta_i = 0$ for $i \neq j$. This interpretation however is extremely unreasonable. Take for example waxed paper for which 13% of the analyzed households bought only one brand. What is it in any one brand that cannot be substituted for by another brand at least to some extent?

The alternative interpretation to the “brand loyal” behavior arises from letting $VS = -1$ and $\theta_i > 0$ for all i , for which $P_{ij} = 0$ and $P_{jj} = 1$. Thus, the “absolute

TABLE 4
Relative Frequency (%) Distributions of Variety Seeking Coefficient (VS)

Product Class	Brand Loyal	Variety Avoiders										Variety Seekers								Average ^d Number of Brands	Zero* Order	Mean ^d VS
		-0.99 -0.51		-0.50 -0.21		-0.20 -0.11		-0.10 -0.01		0	0.10	0.20	0.50	0.99								
		2	9	29	21	17	12	5	3	3	0	269	7.6	33	-0.19							
Facial Tissue ^a	2	9	29	21	17	12	5	3	3	0	269	7.6	33	-0.19								
Aluminum Foil ^a	6	2	25	17	21	4	19	0	2	0	53	4.4	43	-0.15								
Liquid Detergent ^a	1	13	28	16	16	6	8	9	3	0	187	7.1	39	-0.19								
Regular Coffee ^b	20	6	17	10	4	29	4	5	6	0	142	4.0	51	-0.10								
Frozen Unprepared Vegetables ^b	1	3	22	18	12	28	1	5	9	0	219	7.7	53	-0.08								
Waxed Paper ^a	13	11	22	15	2	11	22	4	0	0	46	4.0	43	-0.17								
Ice Cream ^b	4	3	21	13	7	37	1	6	9	0	159	5.5	65	-0.07								
Granulated Sugar ^b	16	0	19	9	2	40	7	2	5	0	43	2.3	60	-0.07								
Bacon ^b	2	2	20	16	10	30	1	4	15	1	168	7.5	60	-0.05								
Butter ^b	4	4	15	16	8	35	2	6	10	0	182	4.9	66	-0.07								
White Bread ^b	1	3	12	14	13	41	4	6	6	0	583	7.7	62	-0.06								
Sausage ^b	2	1	13	14	11	40	4	9	5	0	432	6.5	65	-0.05								
Paper Towels ^b	4	8	13	8	4	44	0	10	10	0	78	4.6	69	-0.07								
Canned Corn ^b	1	1	22	12	7	29	7	9	12	0	69	5.2	65	-0.05								
Dog Food ^b	3	3	9	15	8	39	5	9	9	0	150	8.3	63	-0.03								

Cottage Cheese ^b	9	4	9	9	11	34	6	11	5	2	119	3.9	66	-0.04
Liquid Detergent ^b	11	4	13	8	8	23	4	2	26	2	53	6.8	51	-0.00
Cold Remedies ^a	0	0	2	5	42	7	7	18	18	0	55	8.2	78	0.05
Frozen Concentrated Fruit														
Juices and Drinks ^b	2	4	12	7	10	36	2	7	19	2	129	8.0	65	-0.01
Prepared Cakes ^b	0	0	5	12	7	48	2	6	17	3	193	10.0	70	0.04
Ready to Eat Cereal ^c	0	1	6	10	11	38	2	8	19	5	8376	18.3	60	0.00
Selected Soft Drinks ^b	1	0	4	8	9	45	3	10	19	2	407	11.5	70	0.06
Headache Remedies ^a	3	2	11	8	15	9	23	20	8	3	66	5.2	65	0.03
Canned Juice & Drinks ^b	0	0	3	8	7	47	1	7	23	4	245	12.9	73	0.09
Dry Spaghetti & Macaroni ^b	1	0	2	5	6	34	1	16	33	2	123	7.6	74	0.14
Canned & Frozen Soup ^b	0	1	1	2	5	42	1	8	33	8	332	13.3	67	0.16
Salad Dressing ^b	0	1	2	4	5	29	0	7	46	7	136	9.2	60	0.19
Ready to Eat Cereal ^b	0	0	3	3	3	33	2	9	36	10	317	14.3	55	0.19

^aChicago Tribune Panel, 1958-66.

^bChicago Tribune Panel, 1968.

^cMRCA Panel, 1962-66.

^dExcluding Brand Loyals.

*All households for whom the null hypothesis $V/S = 0$ could not be rejected by the Likelihood Ratio test at $\alpha \leq 0.25$ were included in this category.

loyalty” is in fact an extreme unwillingness to change habits and routines. It seems reasonable therefore to look at the phenomenon of single brand usage as a resentment to change rather than total loyalty to a specific brand. In any case, validation of this construct is called for before a final conclusion can be reached.

The first thing to notice in Table 4 is that in 24 out of the 28 products and in all food products over 50% of the households can be treated as being indifferent to variety. They neither resent it nor actively look for it. Nevertheless, there is still a substantial part of the market that is actively looking for or trying to avoid change. This part is even larger if one includes those consumers for whom VS came out to be insignificantly different from zero.

The second result that is evident from Table 4 is that variety seeking behavior changes across products. In fact the products in Table 4 were ordered according to the skewness of their distribution of VS , from those which have high percentage of variety avoiders ($VS < 0$), through those products for which there is a mixed population of variety avoiders, indifferent and seekers, to those for which relatively large proportion of the consumers prefer varied consumption ($VS > 0$). The results for the different products are not surprising and will probably agree with the intuition of the reader. However, no general underlying constructs could be found to explain the differences among products. It seems that for functional household products where risk of low performance may be an important factor, more variety avoidance is desired. But it is not clear why, for example, the buyers of salad dressing desire more variety than the buyers of ice cream. More will be said on this subject in the next section.

Segmentation by Variety Seeking Behavior

From a managerial point of view the concept of variety seeking is important because different types of variety seeking behavior require different marketing strategies. It would be easier to induce variety seekers to switch to a brand, but harder to keep them from switching away from it. The opposite is true for the variety avoiders. This may imply that a continued short-term promotional activity (e.g. coupons, deals) should be directed at the seekers and massive pulsing strategy is required for avoiders. The introduction of new brands can be much easier in a product class for which there are many variety seekers. In fact the introduction of new brands in such products may be a major marketing tool and a requirement for maintaining market share—ready to eat cereal being a case in point. Clearly then, a successful segmentation of consumers by variety seeking behavior could be very beneficial to management. While it is not the intention of this paper to present complete and detailed segmentation, some indications of the possibilities and difficulties in segmenting the market will be presented.

Segmentation by Demographics

McCann (1974) used MRCA panel data on ready to eat cereal to study the response of different segments of the market to advertising, price and deals. His results reveal significant differences among market segments. Behavioral variables as well as demographics were used to segment the market. In order to explore the possibilities to segment the market by variety seeking behavior three of these variables were chosen for analysis, viz., household income, household size and housewife's age. The same panel data were subjected to the estimation procedure of the variety seeking coefficient of each individual household, as described in previous sections. These results are shown in Table 4. Next, all 8376 households were classified as variety avoiders ($VS < 0$), variety seekers ($VS > 0$) and indifferent ($VS = 0$). Finally, using the same category breakdown that was employed by McCann (1974) the households were cross-classified according to their variety seeking coefficient by income, size and housewife's age. The results are presented in Tables 5 and 6.

TABLE 5
Conditional Probabilities of Level of Variety Seeking by Income, by Age, by Size of Household

Household Size	Age	Household Income*											
		Less Than \$4,000			\$4,000-\$6,999			\$7,000-\$9,999			Over \$10,000		
		V/S < 0	V/S = 0	V/S > 0	V/S < 0	V/S = 0	V/S > 0	V/S < 0	V/S = 0	V/S > 0	V/S < 0	V/S = 0	V/S > 0
1-2	Under 35	0.25	0.47	0.28	0.22	0.63	0.15	0.21	0.61	0.18	0.28	0.58	0.14
	35-55	0.37	0.53	0.10	0.31	0.60	0.09	0.34	0.56	0.10	0.27	0.62	0.11
	Over 55	0.40	0.53	0.07	0.29	0.59	0.12	0.32	0.55	0.13	0.26	0.65	0.09
3	Under 35	0.28	0.61	0.11	0.23	0.58	0.19	0.27	0.58	0.15	0.18	0.56	0.26
	35-55	0.31	0.60	0.09	0.30	0.59	0.11	0.31	0.56	0.13	0.24	0.64	0.12
	Over 55	0.31	0.67	0.03	0.28	0.51	0.21	0.28	0.59	0.13	0.23	0.69	0.08
4-5	Under 35	0.30	0.59	0.11	0.17	0.67	0.16	0.17	0.66	0.17	0.20	0.63	0.17
	35-55	0.24	0.65	0.11	0.25	0.63	0.12	0.20	0.63	0.17	0.22	0.61	0.17
	Over 55	0.36	0.56	0.08	0.24	0.73	0.03	0.31	0.61	0.08	0.29	0.67	0.04
6 +	Under 35	0.15	0.62	0.23	0.11	0.63	0.26	0.08	0.69	0.23	0.04	0.67	0.29
	35-55	0.19	0.73	0.08	0.17	0.67	0.16	0.11	0.64	0.25	0.12	0.64	0.24
	Over 55	0.80	0.20	0.00	0.25	0.63	0.12	0.67	0.33	0.00	0.33	0.67	0.00

* Incomes are in 1962 Dollars.

TABLE 6
*Cross Classification Analysis of Variety Seeking, Income by Age, by Household Size**

Household Size	Age	Household Income			
		Less Than \$4,000	\$4,000 to \$6,999	\$7,000 to \$9,999	Over \$10,000
1 or 2 People	Under 35	0.07 (0.25) 47	0.03 (0.23) 95	0.03 (0.16) 44	-0.03 (0.20) 36
	35-55	-0.04 (0.21) 181	-0.03 (0.18) 282	-0.04 (0.20) 228	-0.01 (0.19) 190
	Over 55	-0.06 (0.19) 1002	-0.02 (0.18) 453	-0.02 (0.18) 199	-0.04 (0.19) 171
	3	0.01 (0.21) 80	0.04 (0.23) 190	0.03 (0.23) 97	0.04 (0.20) 39
	35-55	-0.05 (0.19) 93	-0.02 (0.20) 251	-0.02 (0.18) 224	-0.00 (0.16) 179
	Over 55	-0.07 (0.16) 72	0.01 (0.22) 67	0.01 (0.20) 47	-0.02 0.10 61
4 or 5 People	Under 35	-0.01 (0.17) 150	0.03 (0.17) 576	0.04 (0.16) 380	0.03 (0.15) 134
	35-55	-0.02 (0.15) 104	0.01 (0.17) 450	0.03 (0.15) 453	0.03 (0.18) 361
	Over 55	-0.03 (0.15) 25	-0.03 (0.09) 33	0.00 (0.14) 13	-0.02 (0.07) 24
	More Than 6 People	0.02 (0.20) 53	0.07 (0.18) 221	0.07 (0.16) 137	0.08 (0.15) 69
	35-55	-0.01 (0.18) 62	0.02 (0.15) 189	0.07 (0.20) 173	0.06 (0.17) 147
	Over 55	-0.14 (0.11) 5	-0.06 (0.20) 8	-0.07 (0.06) 3	-0.07 (0.12) 3

*The numbers in each cell correspond to (from top down): Mean *V*S, Standard Error of *V*S, and Number of Households. Incomes are in 1962 Dollars.

Table 5 shows the conditional probability of a household having a certain type of variety seeking behavior given its segment identity. Although in all segments the indifferent group is dominant, as it is for the whole sample (60%), some clear differences among the segments do exist. In all cases the probability of a household with a wife under 35 years old to be a variety seeker is higher than that of households with an older wife. In the low income (less than \$4000), small household size (1–2) this probability for young households is four times that of an old household (0.28/0.07). Likewise, the probability of being a variety avoider is higher for old households than for young ones. The effect of income seems to interact with the household size in such a way that households with average per capita income tend to be more variety seekers than others. Table 6 gives mean *VS* for each segment as well as the standard deviation of *VS* and the segment size. Again, it seems that young households with average per-capita income tend to be more variety seekers than others, and old people tend to be more conservative in their purchase behavior than others. However, no strong conclusions can be drawn from Table 6 since the differences among the groups' means are not large relative to their standard deviations (within segment variation); still the directions are clear. It should be kept in mind that this segmentation is not necessarily the best possible in terms of separating seekers from avoiders but one that was found to be useful by McCann (1974).

Segmentation Across Products

In the previous section some positive indications were given for the possibility of beneficial segmentation by variety seeking behavior. This possibility brings up the issue of the stability of these segments across product classes. If households' behavior does not change much from one product to another, or if the changes follow a certain identifiable systematic pattern, then the results of segmentation studies can be carried over from one product class to another. To investigate this possibility the estimated *VS* coefficients for the households in the sample were correlated across the 21 products for which data from the same panel (Chicago Tribune, 1968) were available. Correlation coefficients were in the range of ± 0.5 , but most of them were around zero and no underlying structure could be found by factor analyzing these correlations. This result indicates that households' behavior does change among products, and a specifically designed research may be necessary to identify the pattern of change.

Limitations and Extensions

The variety seeking model presented in equations (5) and (6) is a very simplistic and crude one. It was put forward as a first attempt to empirically estimate the extent of this phenomenon. Its main advantage is the relative ease with which it can be estimated. However, it ignores three important aspects of variety seeking, namely time, cumulative experience and brand attributes.

The shorter the inter-purchase time the more important is the need for variety that arises from boredom and satiation. Similarly, boredom and satiation are induced by cumulative experience with the same brand. The model that was presented is a first order stochastic process that ignores this effect by considering only the last purchase occasion. Jeuland (1978) has formulated a variety seeking model which incorporates inter-purchase time as well as cumulative experience. Unfortunately, the number of parameters needed to be estimated and the complicated functional form render the model impractical for empirical work.

It seems that the most restrictive feature of the variety seeking model in (5) and (6), as well as the Jeuland's model, is the assumption that the variety gained by switching between brands does not depend on the brands involved. While this assumption may be adequate for highly similar and visible products, it is intuitively appealing to assume

that in many products consumers seek variety among product attributes rather than brand names. McAlister (1982) captured the aspects of time, cumulative experience and brand attributes, by using the concept of attribute satiation to link accumulated attributes inventories over time with varied consumption. This model is an excellent tool for validation of variety seeking constructs. However, its data requirements render it impractical for an extensive empirical investigation of variety seeking in different markets.

The simple model in equations (5) and (6) can be modified to incorporate the effect of brand attributes on varied consumption. Assume that the consumer partitions the brands in a product category according to some main attributes and satisfies her need for variety by switching among partitions. For such a consumer the probability of switching from brand i in partition k to brand j in another partition can be described by

$$P_{ij} = \frac{(|VP| + VP)\theta_j}{2(1 - \Pi_k)} + (1 - |VP|)\theta_j \quad \text{where} \quad (9)$$

$$-1 \leq VP \leq 1, \quad i \in k, \quad j \notin k, \quad \Pi_k = \sum_{i \in k} \theta_i.$$

In this modified model the variety seeking force ($VP > 0$) is weighted by the preference of the brand *to* which the consumer switches relative to the summed preferences of all the brands that are not in the partition of the brand *from* which the consumer switches. This is in contrast to the basic model in (5) which gives equal weights ($1/(n-1)$) to all brands other than the one from which the consumer switches. The modified model weighs each partition by its summed preference (Π) relative to the preferences of all partitions competing for a switch out of partition k , $(1 - \Pi_k)$. "Variety avoidance" in this partitioned market means staying in the same partition, but not necessarily with the same brand. It is assumed that no variety is gained by switching within the same partition.

Thus, the probability of switching to a brand within the same partition is relative to its preference weight in its own partition,

$$P_{ij} = \frac{(|VP| - VP)\theta_j}{2\Pi_k} + (1 - |VP|)\theta_j, \quad i \in k, \quad j \in k. \quad (10)$$

The two models in (5)–(6) and (9)–(10) were applied to the laboratory experimental data which resulted from the doctoral dissertation work of Reibstein (1975). 138 student and secretarial subjects participated in the experiment daily for six weeks, a total of 30 trials. On each trial the subjects had to choose one out of six brands of carbonated beverage. For the application of the modified model in (9)–(10) the six brands were partitioned to colas (four brands) and un-colas (two brands). The results for the two models are summarized in Table 7.

The first and second rows of Table 7 give the results for the two models separately. The third row gives the percentage of consumers for whom the basic model (VS) showed a better fit. The fourth row gives the percentage of consumers for whom variety seeking between partitions fitted better. Finally, the last row gives the distribution of variety seeking of either model.

Ten percent of the 138 subjects chose the same brand throughout the 30 trials. Eleven percent were found to be best described by the zero order process, i.e., $VS = VP = 0$. Sixty percent were described best by the partitions model (VP) while only 19 percent were better described by variety seeking among brands (VS) rather

TABLE 7
Relative Frequency (%) Distributions of Variety Seeking Coefficients*

	- 0.99	- 0.50	- 0.20	0	0.01	0.21	0.51	Brand		
	- 0.51	- 0.21	- 0.01		0.20	0.50	0.99	Loyal	Total	Zero**
<i>VS</i>	2	5	7	45	15	13	3	10	100	66
<i>VP</i>	7	6	12	18	37	9	1	10	100	50
<i>VS</i> Consumers	0	4	2	—	7	4	2	—	19	7
<i>VP</i> Consumers	3	5	10	—	31	9	2	—	60	33
Either Model	3	9	12	11	38	13	4	10	100	40

* All figures are percents out of 138 consumers.
 **All subjects for whom the variety seeking coefficients were not significantly different from zero at $\alpha \leq 0.25$.

than partitions. These subjects may have had another partitioning in mind, regular versus diet for example. Overall 24 percent of the subjects were “Inert” (*VS* or *VP* smaller than zero), and 55 percent were variety seekers (*VS* or *VP* larger than zero). By applying the *VP* model to panel data and trying different partitionings, the manufacturer of the product can get a detailed description of his market and plan his brands line accordingly.

The results that were reported in Table 4 come from actual market data, as they were reported by a panel of consumers. Thus, the actual switching among brands reflects the effects of competitive actions in those markets as well as the effect of variety seeking. Since the model suggested here does not incorporate marketing mix variables, it would not be reasonable to expect it to be a good predictor of brand switching in an actual market situation. However, the model can still be useful for identifying patterns of behavior by comparison to models that have different implications on brand switching behavior. The test against the zero order model that was reported in Tables 4 and 7 is one such comparison.

The availability of the soft drinks experimental data offers a unique opportunity to test the goodness of fit of the model, since no marketing mix effects are present in the data. In order to perform this test a six by six transition matrix was computed for each one of the 138 subjects who participated in the experiment. An element of this matrix, n_{ij} , is the number of times the subject switched from brand i to brand j . This observed matrix should be compared to an expected switching matrix predicted by the model. Since the expected matrix for a typical subject will have many sparse values, the usual chi-square test is not appropriate. Instead, a test statistic suggested by Shields and Heeler (1979) was used,

$$\Lambda_s = 2 \left[\sum_{ij} \ln(n_{ij}!) - \sum_{ij} \ln(m_{ij}!) \right],$$

where n_{ij} is as defined above and m_{ij} is the expected number of switches from brand i to brand j , rounded to the closest integer, $m_{ij} = n_i \cdot P_{ij}$, where P_{ij} is as defined in (9) and (10), and $n_i = \sum_{j=1}^6 n_{ij}$ is the total number of switches out of brand i . The P_{ij} 's were computed by using the maximum likelihood estimates of the parameters θ_j 's and *VP*. The statistic Λ_s is distributed asymptotically as chi-square. To get the degrees of freedom for this chi-square statistic, one should subtract from the size of the matrix ($6 \times 6 = 36$) six degrees of freedom for the row totals, n_i 's, six additional degrees of freedom are subtracted for the estimated parameters of the model. One degree of freedom is lost for every two cells (m_{ij} 's) with values of zero or one (see Shields and Heeler 1979).

The Λ_s statistic was computed for each of the 124 subjects who chose more than one brand in the 30 trials. All but 14 subjects had Λ_s values not significant at a significance

level of 0.50. Of these 14 subjects, 8 had Λ_y values not significant at 0.25 level, 4 were not significant at 0.10, and 2 were not significant at 0.05 level. It can be concluded, therefore, that for these experimental data the *VP* model had a remarkable fit.

Summary and Conclusions

This paper ties together the phenomena of brand switching and variety seeking. Brand switching and repeat buying are shown to be a combined result of underlying brand preferences and consumer's preference for change. The suggested variety seeking model offers new interpretation and explanation to the phenomena of "absolute brand loyalty" and "zero order." "Absolute brand loyalty" is reinterpreted as a strong resistance to change. The "zero order—no feedback effect" model is seen here as inactivity with regard to change, which implies either indifference to change or satisfaction with the variety brought about by "other factors" or perhaps by chance.

The empirical results indicate that in food items many of the consumers are not actively seeking to increase or avoid variety in their consumption. However, there is a substantial part of the market that does not behave indifferently toward variety and should not be ignored. For some nonfood products this part of the market is dominant. These are the more functional and less conspicuous products. Products were shown to vary with respect to the proportions of variety seekers and avoiders, but no underlying explanations could be found for these differences. The benefits of market segmentation by variety seeking were pointed out and some empirical evidence was presented to demonstrate the possibilities and difficulties of such segmentation.

In a section that deals with possible extensions, the basic model was modified to deal with product market partitioning. The empirical results show better fit of this model.

This study raises some unresolved issues for further research. First, the variety seeking concept requires empirical validation. This concept is not undimensional. Different measures can and should be developed and correlated with the one advanced in this paper. Such validation may resolve the issue which was raised above with regard to the interpretation of the zero order process. Second, more research is needed to study the characteristics of variety seekers, perhaps along the lines of past research on innovativeness and risk taking. Finally, a cross-sectional study of products may prove to be very helpful in understanding the phenomenon of variety seeking.

Appendix. Power of the Test

For the likelihood ratio test that $VS = 0$ the asymptotic power can be computed for given α and individual purchase sequence. The null (H_0) and the alternative (H_1) hypotheses for this test are $H_0: VS = VS_0 = 0$ and $H_1: VS \neq VS_0$. In order to compute the power a simple alternative hypothesis should be specified, $H_1: VS = VS_1$. Kendall and Stuart (1961, pp. 230–231) show that under the alternative hypothesis $-2 \log(F_0/F_1)$ is asymptotically distributed as a noncentral χ^2 with one degree of freedom and a noncentral parameter $\lambda = (VS_1 - VS_0)^2 V^{-1}$, where

F_0 is the likelihood function under H_0 as in (7c),

F_1 is the likelihood function under H_1 as in (7a) or (7b) according to whether $VS_1 > 0$ or $VS_1 < 0$, respectively, and

$$V^{-1} = -E\left(\frac{\partial^2 \log F_1}{\partial VS^2}\right)$$

is the inverse of the asymptotic variance of the maximum likelihood estimator of VS . The θ_i 's in F_1 should be replaced by their estimates given in (8a) and (8b).

For $VS_0 = 0$ and $VS_1 > 0$ the noncentral parameter turns out to be

$$\lambda = VS_1^2 \left\{ \frac{N-1}{(1-VS_1)} + (N-1)(n-1) \sum_i \frac{[1/(n-1) - \hat{\theta}_i]^2}{VS_1/(n-1) + (1-VS_1)\hat{\theta}_i} + \frac{[1/(n-1) - \hat{\theta}_1]^2}{[VS_1/(n-1) + (1-VS_1)\hat{\theta}_1]^2} - \frac{1}{(n-1+VS)^2} \right\}.$$

For $VS_0 = 0$ and $VS_1 < 0$ the noncentral parameter is

$$\lambda = VS_1^2(N-1) \left\{ \frac{(n-1)}{(1+VS_1)} + \sum_i \frac{(1-\hat{\theta}_i)^2}{(1+VS_1)\hat{\theta}_i - VS_1} \right\}.$$

Since $-2 \log[F_0/F_1]$ has a noncentral χ^2 distribution with *one* degree of freedom the standard normal distribution can be used to compute β (see Kendall and Stuart 1961, pp. 231–233) as follows

$$\beta = \Pr(-Z_{\alpha/2} - \lambda^{1/2} \leq z \leq Z_{\alpha/2} - \lambda^{1/2}),$$

where z is the standard normal deviate and $\alpha = \int_{-\infty}^{\infty} f(z) dz$.⁵

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